

ASSIGNMENT – 1

Applications of Deep Learning in Full Stack Web Development

1. Introduction

Deep Learning is a branch of Artificial Intelligence that enables computers to learn from large amounts of data. In Full Stack Web Development, deep learning is integrated with the frontend, backend, databases, and APIs to build intelligent web applications.

2. Top 10 Applications

No.	Application	Purpose
1	AI Chatbot	Automatic customer support
2	Image Recognition	Object and face detection
3	Recommendation System	Product & movie recommendations
4	Speech Recognition	Voice-to-text conversion
5	Text Classification	Spam & sentiment analysis
6	Face Authentication	Secure login
7	OCR	Extract text from images
8	Medical Diagnosis	Disease detection from scans
9	Fraud Detection	Detect suspicious transactions
10	Language Translation	Translate languages

3. Why We Chose AI Chatbot

AI Chatbot is one of the best examples of deep learning in full stack web development. It combines frontend, backend, databases, and AI models to provide intelligent conversations. It is widely used in education, banking, healthcare, e-commerce, and customer support.

4. Advantages

- 24/7 availability
- Instant responses
- Improves customer experience
- Reduces human effort
- Saves time and cost
- Handles multiple users.

5. Technologies Used

Frontend: HTML, CSS, JavaScript, React

Backend: Node.js & Express

Deep Learning: TensorFlow, PyTorch, Keras, LLMs

Database: MySQL / MongoDB

API: REST API

6. AI Chatbot Architecture

User → Frontend (React) → Backend (Node.js) → Deep Learning Model → Database (Optional) → Response to User

7. Conclusion

AI Chatbot demonstrates how deep learning can make full stack web applications intelligent, efficient, and user-friendly.

8. References

1. TensorFlow Documentation
2. PyTorch Documentation
3. React Documentation
4. Node.js Documentation